

LAB EXPERIMENT RECORD

SET 5: Website Analytics

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Thursday

Aim

To analyze website traffic data using different data visualization techniques such as line chart, bar chart, stacked area chart, and an interactive Tableau dashboard.

Dataset: Website Traffic

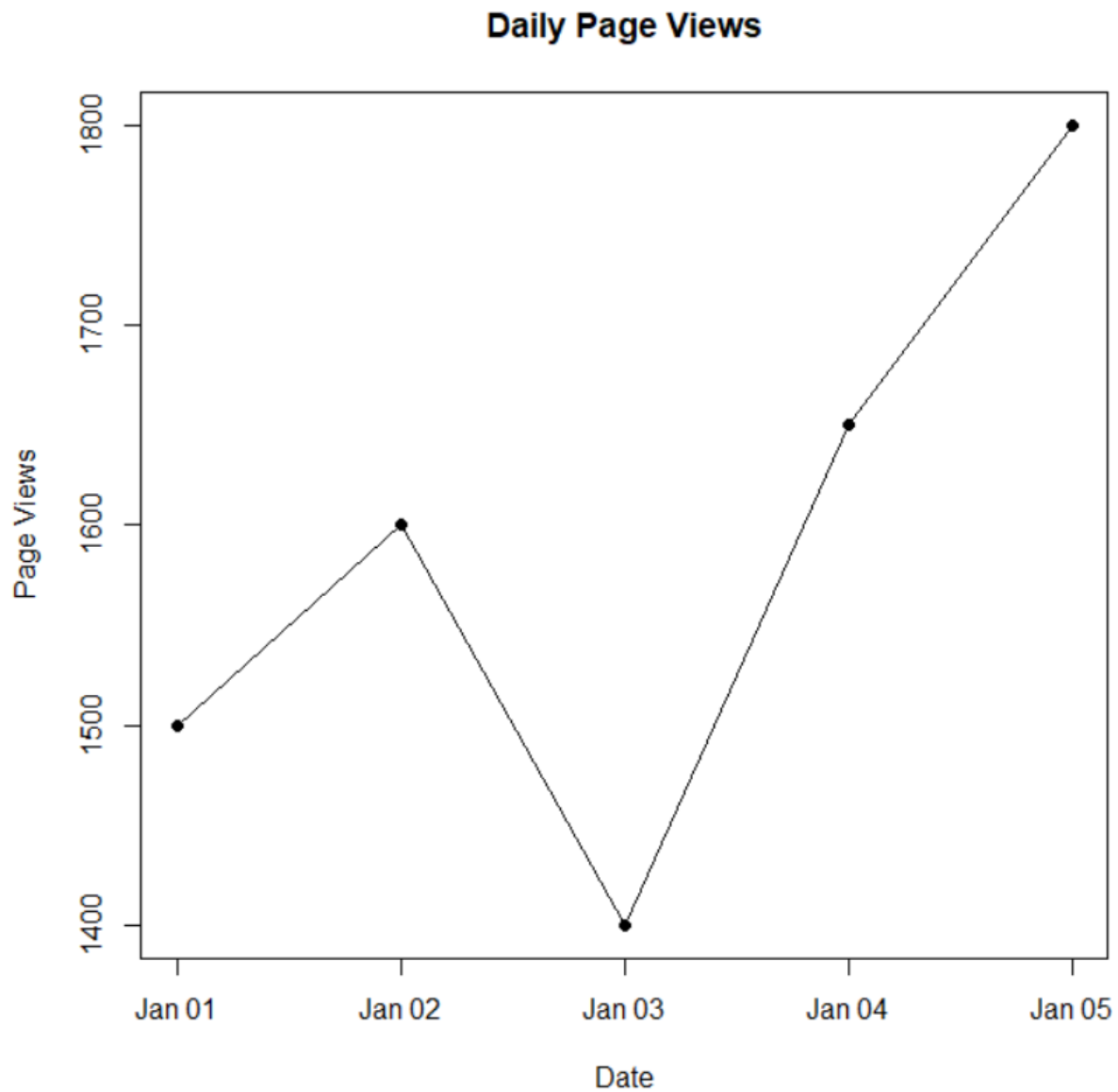
Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

Note: The original dataset does not provide **Likes, Shares, and Comments**. Sample values are therefore used only for Program 3 because those variables are required by the question.

Program 1: Daily Page Views Line Chart

Code

```
date <- as.Date(c("2023-01-01",  
                  "2023-01-02",  
                  "2023-01-03",  
                  "2023-01-04",  
                  "2023-01-05"))  
  
page_views <- c(1500, 1600, 1400, 1650, 1800)  
  
website <- data.frame(date, page_views)  
  
plot(website$date,  
      website$page_views,  
      type = "o",  
      main = "Daily Page Views",  
      xlab = "Date",  
      ylab = "Page Views",  
      pch = 19)
```



Observation

The page views generally increased during the five-day period. The highest number of page views was **1800 on January 5**, while the lowest was **1400 on January 3**.

Program 2: Top 3 Days by Click-through Rate

Code

```
date <- as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03",
                  "2023-01-04",
                  "2023-01-05"))

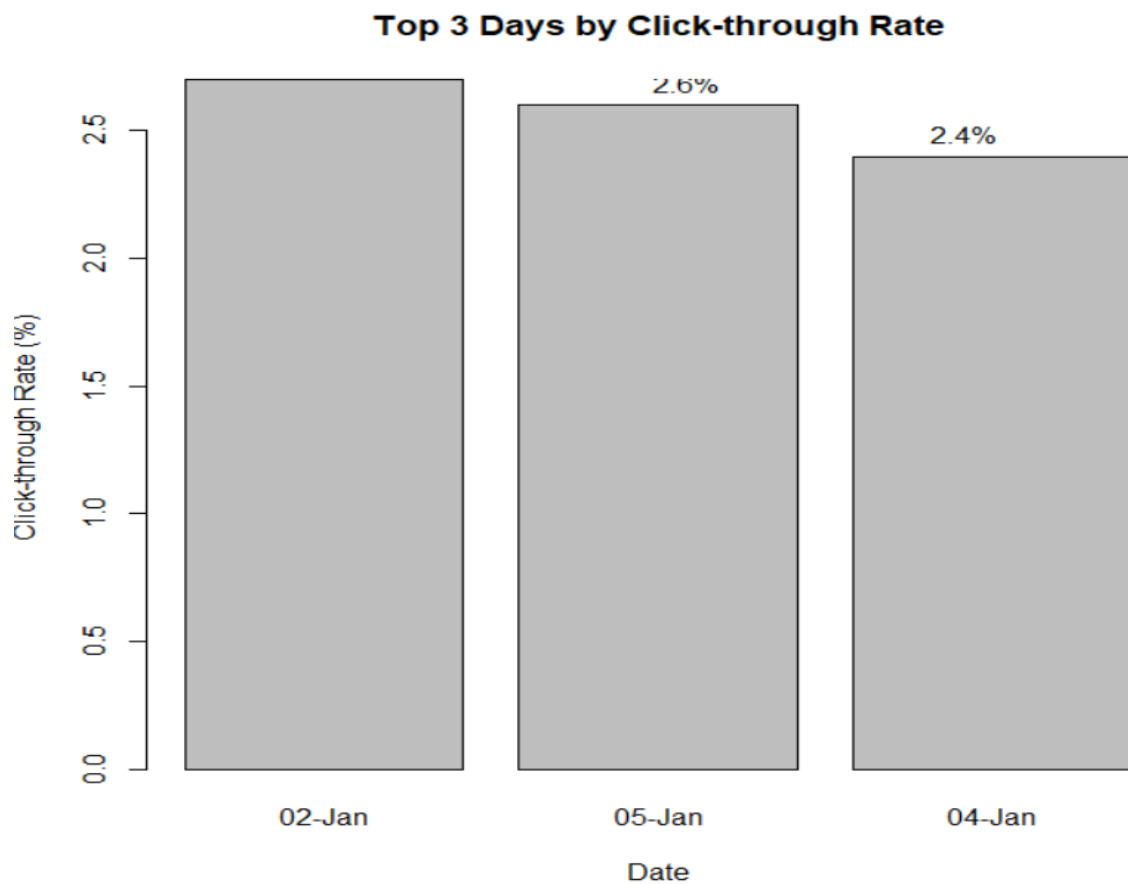
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

website <- data.frame(date, ctr)

top_days <- website[
  order(-website$ctr),
  ][1:3, ]

barplot(top_days$ctr,
        names.arg = format(top_days$date, "%d-%b"),
        main = "Top 3 Days by Click-through Rate",
        xlab = "Date",
        ylab = "Click-through Rate (%)")

text(seq_along(top_days$ctr),
     top_days$ctr,
     labels = paste0(top_days$ctr, "%"),
     pos = 3)
```



Observation

The top three days according to click-through rate are:

Date	Click-through Rate
January 2	2.7%
January 5	2.6%
January 4	2.4%

January 2 has the highest click-through rate of **2.7%**.

Program 3: Stacked Area Chart – User Interactions

Sample Data

Date	Likes	Shares	Comments
2023-01-01	300	100	50
2023-01-02	350	120	60
2023-01-03	280	90	45
2023-01-04	400	130	70
2023-01-05	450	150	80

Code

```
date <- as.Date(c("2023-01-01",  
                  "2023-01-02",  
                  "2023-01-03",  
                  "2023-01-04",  
                  "2023-01-05"))
```

```
likes <- c(300, 350, 280, 400, 450)  
shares <- c(100, 120, 90, 130, 150)  
comments <- c(50, 60, 45, 70, 80)
```

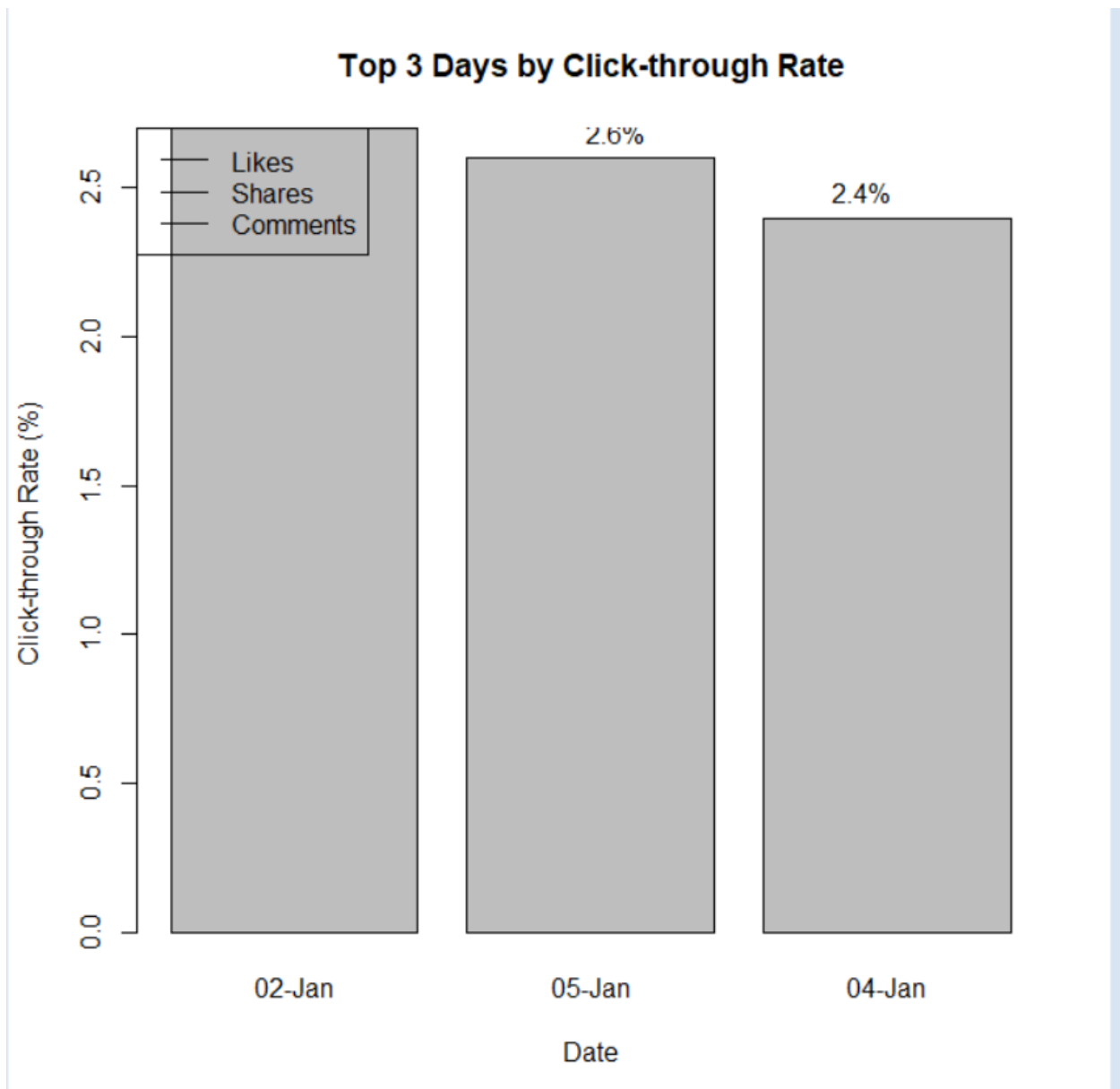
```
interaction <- data.frame(  
  date, likes, shares, comments  
)
```

```
interaction_matrix <- t(  
  as.matrix(  
    interaction[, c("likes",  
                    "shares",  
                    "comments")]
```

```
)  
)
```

```
matplot(  
  interaction$date,  
  interaction_matrix,  
  type = "l",  
  lty = 1,  
  lwd = 2,  
  main = "User Interactions Over Time",  
  xlab = "Date",  
  ylab = "Number of Interactions"  
)
```

```
legend(  
  "topleft",  
  legend = c("Likes", "Shares", "Comments"),  
  lty = 1  
)
```



Observation

Likes have the highest number of interactions throughout the period. Shares and comments have comparatively lower values. Overall user interactions increase toward January 5.

Program 4: Tableau Dashboard

For this program, use **Tableau**, because the original question specifically asks for a Tableau dashboard.

Dashboard Components

Sheet 1 – Line Chart

- Drag Date → Columns
- Drag Page Views → Rows
- Select **Line Chart**
- Add title: **Daily Page Views**

Sheet 2 – Bar Chart

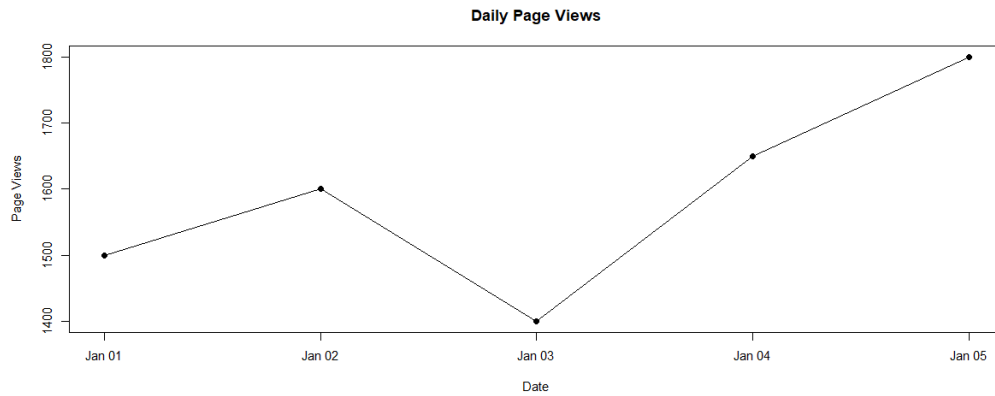
- Drag Date → Columns
- Drag Click-through Rate → Rows
- Select **Bar Chart**
- Sort in descending order
- Add data labels
- Add title: **Click-through Rate by Date**

Dashboard

1. Open New **Dashboard**.
2. Drag the **Daily Page Views Line Chart** onto the dashboard.
3. Drag the **Click-through Rate Bar Chart** onto the dashboard.
4. Add suitable filters if required.
5. Add the title **Website Analytics Dashboard**.
6. Use the dashboard to interactively explore website traffic.

Website Analytics Dashboard

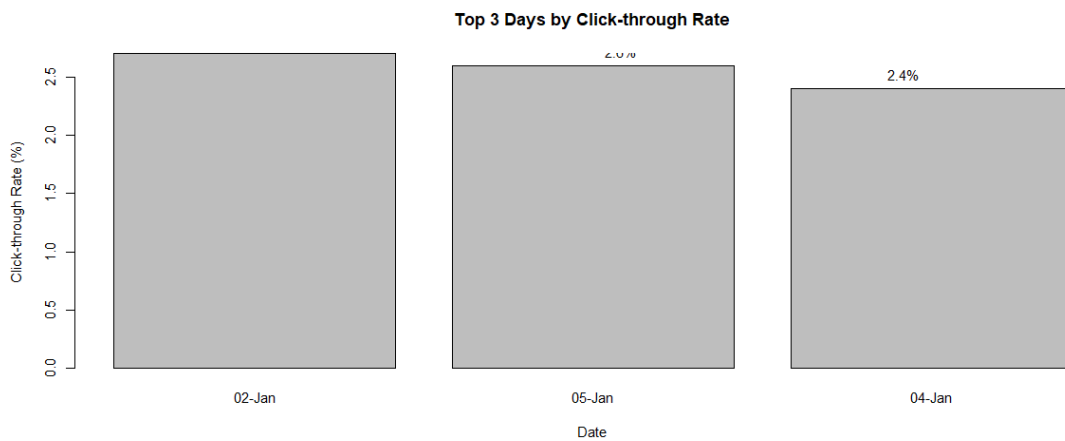
Daily Page Views



Top 3 Days by Click-through Rate



Top 3 Days by Click-through Rate



Observation

The Tableau dashboard combines the page views line chart and click-through rate bar chart, allowing users to compare website traffic trends and click-through rates interactively.

Procedure

1. Enter the website traffic dataset.
2. Convert the date values into Date format.
3. Create a line chart to visualize daily page views.
4. Create a bar chart to identify the top three days based on click-through rate.
5. Add the sample interaction dataset containing likes, shares, and comments.
6. Create a stacked area-style visualization to analyze user interactions.
7. Import the website traffic data into Tableau.
8. Create the page views line chart in Tableau.
9. Create the click-through rate bar chart in Tableau.
10. Combine both charts into a single interactive dashboard.
11. Add appropriate titles, labels, and filters.

Result

The website traffic data was successfully analyzed using **line chart, bar chart, and stacked interaction visualization** techniques. An interactive **Tableau dashboard** was also developed by combining the line chart and bar chart for website traffic exploration.

Key R Functions Used

R Function	Purpose
c()	Creates vectors
as.Date()	Converts values into date format
data.frame()	Creates data frames
plot()	Creates line charts
barplot()	Creates bar charts
order()	Sorts data
format()	Formats dates
text()	Adds labels to charts

matplot()	Creates multi-series plots
as.matrix()	Converts data into matrix format
t()	Transposes a matrix
legend()	Adds chart legends